EXPERIMENT-5

Implement the K-Nearest Neighbors (KNN) algorithm in Python.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix

print("="*70)
print("          K-NEAREST NEIGHBORS (KNN)")
print("="*70)

uploaded = files.upload()
data = pd.read_csv("Iris.csv")

print("\nDataset\n")
print(data.head(10))

X = data.iloc[:,1:5]
y = data.iloc[:,5]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train,X_test,y_train,y_test=train_test_split(
X,y,test_size=0.3,random_state=42)
```

```
print("\nTraining Samples :",len(X_train))
print("Testing Samples :",len(X_test))

model=KNeighborsClassifier(n_neighbors=3)

print("\nTraining KNN Model...")

model.fit(X_train,y_train)

prediction=model.predict(X_test)

print("\nActual Values")
print(y_test)
print("\nPredicted Values")
print(prediction)
print("\nAccuracy")
print(round(accuracy_score(y_test,prediction)*100,2),"%")
print("\nConfusion Matrix")
print(confusion_matrix(y_test,prediction))

sample=[[5.1,3.5,1.4,0.2]]
result=model.predict(sample)
print("\nNew Sample :",sample)
print("Predicted Class :",encoder.inverse_transform(result))
```

Sample Input:

```
======================================================================
          K-NEAREST NEIGHBORS (KNN)
======================================================================
Choose Files Iris.csv
Iris.csv(text/csv) - 5107 bytes, last modified: 8/5/2026 - 100% done
Saving Iris.csv to Iris (1).csv
```

Sample Output:

```
...  Dataset

      Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Species
0      1            5.1           3.5            1.4           0.2  Iris-setosa
1      2            4.9           3.0            1.4           0.2  Iris-setosa
2      3            4.7           3.2            1.3           0.2  Iris-setosa
3      4            4.6           3.1            1.5           0.2  Iris-setosa
4      5            5.0           3.6            1.4           0.2  Iris-setosa
5      6            5.4           3.9            1.7           0.4  Iris-setosa
6      7            4.6           3.4            1.4           0.3  Iris-setosa
7      8            5.0           3.4            1.5           0.2  Iris-setosa
8      9            4.4           2.9            1.4           0.2  Iris-setosa
9     10            4.9           3.1            1.5           0.1  Iris-setosa

Training Samples : 105
Testing Samples : 45

Training KNN Model...

Actual Values
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Values
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy
100.0 %

Confusion Matrix
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

New Sample : [[5.1, 3.5, 1.4, 0.2]]
Predicted Class : ['Iris-setosa']
```